Exploratory Data Analysis in R

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KOHL CENTRE

What is EDA?

> EDA is the practice of **systematically cleaning and understanding your data** before you build any model or write any finding.

EDA has two jobs:

1. **Data cleaning** — find and fix problems: missing values, impossible values, coding errors, logical inconsistencies
2. **Understand your data** — see the shape of distributions, spot outliers, understand relationships among variables

Today's 6-step workflow:

Step	What you do
1	Summary statistics
2	Spot problems in the numbers
3	Hidden missing values
4	Anomalies and outliers
5	Distribution shape
6	Correlations

Step 1 — Summary Statistics

`skimr::skim()` — One Function, Full Picture

```
library(skimr)

penguins |> skim()
```

Data summary

Name	penguins
Number of rows	344
Number of columns	8
_____	
Column type frequency:	
factor	3
numeric	5
_____	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
species	0	1.00	FALSE	3	Ade: 152, Gen: 124, Chi: 68
island	0	1.00	FALSE	3	Bis: 168, Dre: 124, Tor: 52
sex	11	0.97	FALSE	2	mal: 168, fem: 165

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
bill_length_mm	2	0.99	43.92	5.46	32.1	39.23	44.45	48.5	59.6	
bill_depth_mm	2	0.99	17.15	1.97	13.1	15.60	17.30	18.7	21.5	
flipper_length_mm	2	0.99	200.92	14.06	172.0	190.00	197.00	213.0	231.0	
body_mass_g	2	0.99	4201.75	801.95	2700.0	3550.00	4050.00	4750.0	6300.0	
year	0	1.00	2008.03	0.82	2007.0	2007.00	2008.00	2009.0	2009.0	

→ Open `R/demos/dataviz02/01-eda-workflow.R` to run all 6 EDA steps interactively — from `skim()` through correlation heatmaps.

Reading the `skim` Output — What to Check First

Columns to look at immediately:

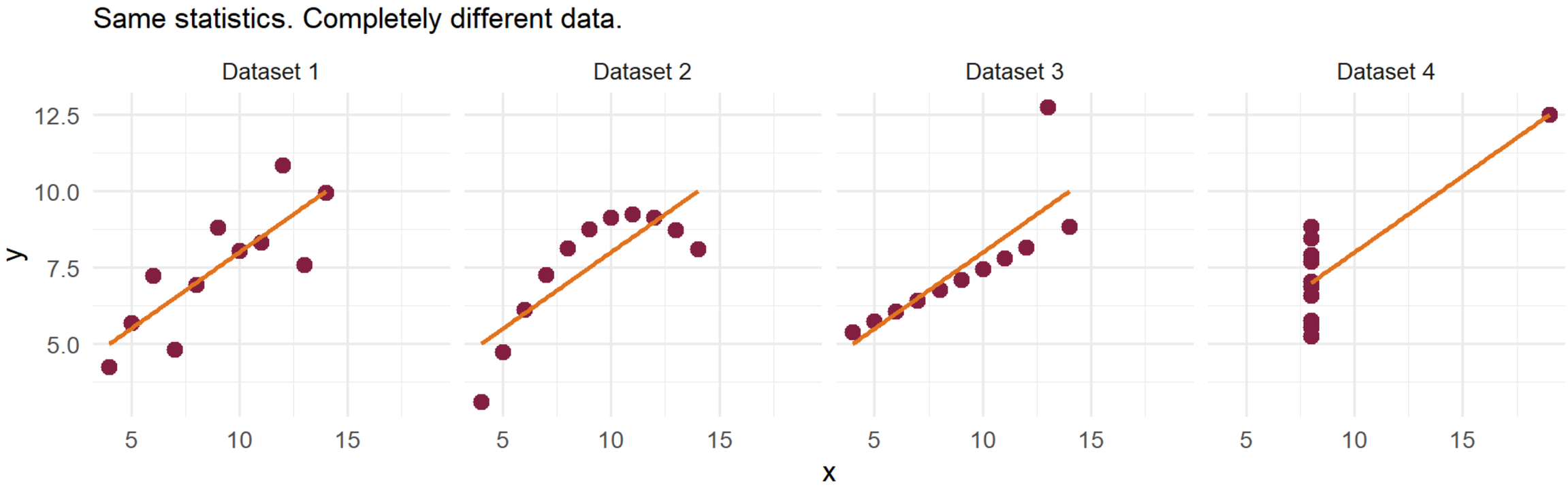
Column	What to check
`n_missing`	Any variable with many missing?
`complete_rate`	high → investigate
`numeric.min`	Can this value realistically be this low?
`numeric.max`	Percent > 100%? Count > total population?
`numeric.mean`	Does this match your expectations?
`numeric.hist`	Very skewed? Long tail?

But `skim()` Is Not Enough

Consider these four datasets — all with nearly identical summary statistics:

dataset	mean(x)	mean(y)	sd(x)	sd(y)	r(x, y)
Dataset 1	9	7.5	3.32	2.03	0.816
Dataset 2	9	7.5	3.32	2.03	0.816
Dataset 3	9	7.5	3.32	2.03	0.816
Dataset 4	9	7.5	3.32	2.03	0.817

The Reveal — Always Plot



The numbers said they were identical. The plots tell four different stories.

> Summary statistics are a starting point, not a conclusion. Always follow `skim()` with visualizations of your key variables.

Step 2 — Spot Problems in the Numbers

High Missingness and Impossible Values

```
# After skim(), drill into specific concerns

# 1. Which variables have meaningful missingness?
penguins |>
  summarise(across(everything(), ~ mean(is.na(.x)))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "pct_missing") |>
  filter(pct_missing > 0) |>
  arrange(desc(pct_missing))

# 2. Check value ranges — should never be negative or > 100
state_data |>
  filter(pct_college < 0 | pct_college > 100) |>
  select(NAME, pct_college)

# 3. Check for impossible values specific to your data
penguins |>
  filter(body_mass_g < 0 | bill_length_mm > 200)
```

Logical Consistency Checks

Some problems only appear when you cross-check variables against each other:

```
# Pattern: components should sum to the total
# Example with ACS college attainment (male + female = total)
acs_wide |>
  mutate(
    pct_recalculated = (col_male + col_female) / pop_total * 100,
    discrepancy      = abs(pct_recalculated - pct_college)
  ) |>
  filter(discrepancy > 1) |>          # flag > 1 percentage point off
  select(NAME, pct_college, pct_recalculated, discrepancy)

# Other common checks in public data:
# - percentage columns that don't sum to 100% across categories
# - end_date earlier than start_date
# - children_count > household_size
# - mortality_rate > 1 (if expressed as fraction not per 100k)
```

One sentence rule: For every derived variable you create, write the formula and check that it holds for all rows.

Step 3 — Hidden Missing Values

Explicit vs Implicit Missing

```r
stocks <- tibble(
  year  = c(2020, 2020, 2020, 2020, 2021, 2021, 2021),
  qtr   = c(   1,    2,    3,    4,    2,    3,    4),
  price = c(1.88, 0.59, 0.35,   NA, 0.92, 0.17, 2.66)
)
stocks |> complete(year, qtr)
```

```
# A tibble: 8 × 3
   year   qtr price
  <dbl> <dbl> <dbl>
1  2020     1  1.88
2  2020     2  0.59
3  2020     3  0.35
4  2020     4 NA
5  2021     1 NA
6  2021     2  0.92
7  2021     3  0.17
8  2021     4  2.66
```

- Q4 2020 price is **explicitly missing** — the `NA` is present in the data
- Q1 2021 is **implicitly missing** — the row never appears at all
- `complete(year, qtr)` converts implicit → explicit, making gaps visible

Identifying Hidden Missing Values

Not all missing data shows up as `NA`. Two patterns to watch for:

Sentinel / placeholder codes — numeric values used to mean "no data":

Code	Common context
`0`	Counts that should never be zero (e.g., farm acreage)
`6`, `9`	Legacy survey codes for "not applicable" or "refused"
`99`, `-99`	Short numeric fields where 99 means "unknown"
`-999`, `9999`	Longer fields — check data documentation

```
# Flag and replace sentinel values before analysis
df |> mutate(income = na_if(income, 9999),
             age    = na_if(age, 99))
```

Logically Missing Values

Logically missing — absence driven by the real world, not data error:

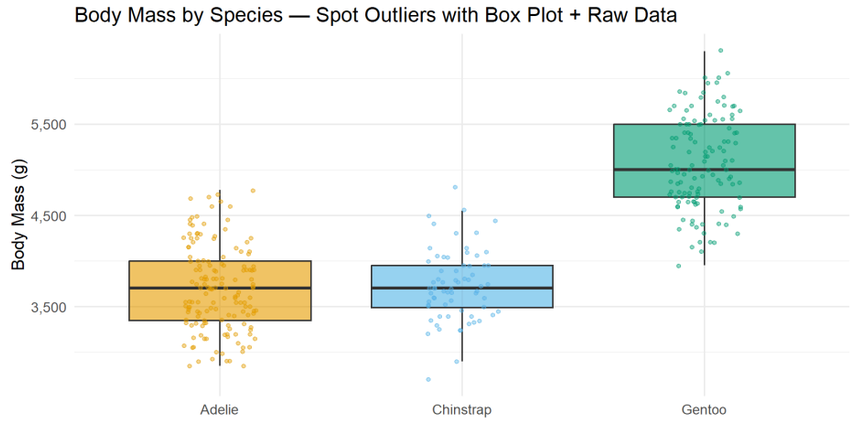
- Wage data is missing for unemployed individuals — *because* they have no wage, not at random
- **Confidential suppression** — USDA agricultural reports omit data when too few farms report, to protect individual privacy; the absence signals "too few observations," not zero
- Test scores absent for students who dropped out — selection, not random dropout

> Always ask: *why* is this value absent? The reason shapes how you handle it.

Step 4 — Anomalies and Outliers

Box Plot + Raw Data — Spot the Outliers

```
penguins |>
  ggplot(aes(x = species, y = body_mass_g, fill = species)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_jitter(aes(color = species), width = 0.15, alpha = 0.4, size = 1.2) +
  scale_fill_manual(values  = c("#E69F00", "#56B4E9", "#009E73")) +
  scale_color_manual(values = c("#E69F00", "#56B4E9", "#009E73")) +
  scale_y_continuous(labels = label_comma()) +
  labs(title = "Body Mass by Species — Spot Outliers with Box Plot + Raw Data",
       x = NULL, y = "Body Mass (g)") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "none")
```



`outlier.shape = NA` suppresses duplicate outlier dots from the boxplot — they are already shown by `geom_jitter`. Investigate any points well beyond the whiskers before dropping them.

What to Do With an Outlier

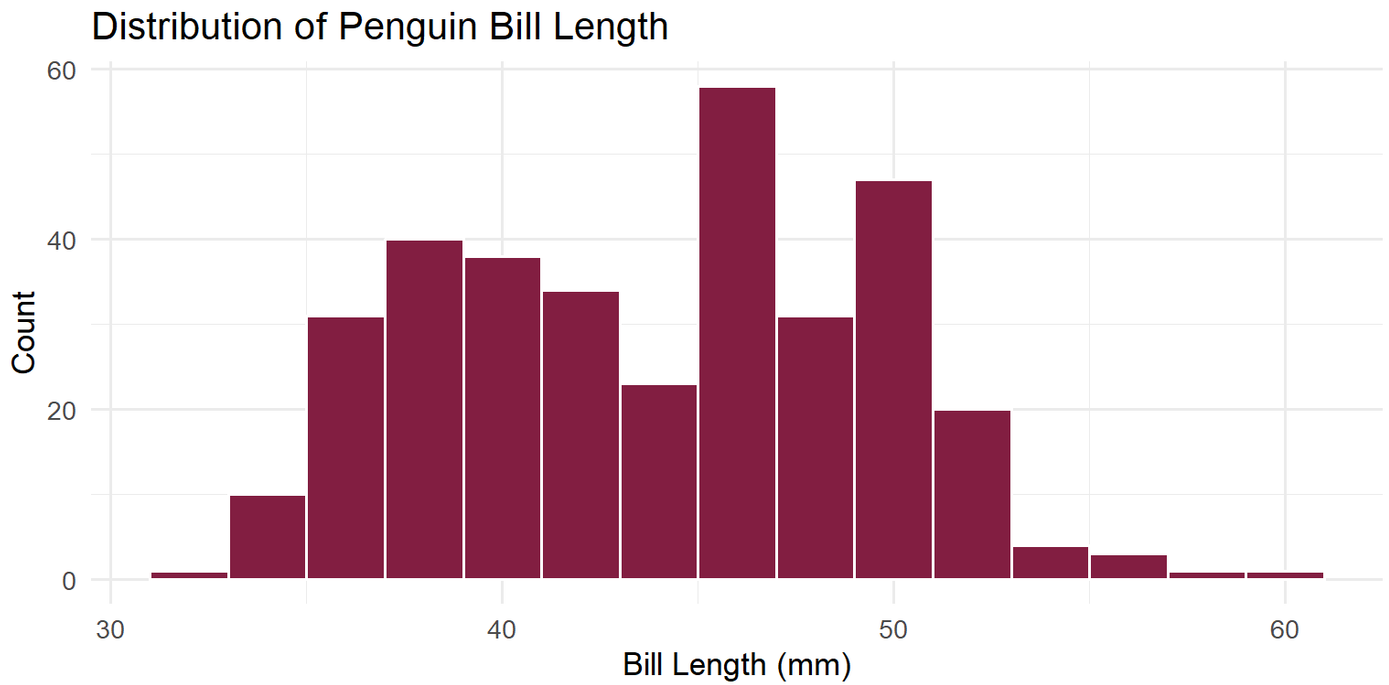
An outlier is not automatically an error. It falls into one of three categories:

1. **Unknown data error** — a typo, impossible value, or merge mistake → investigate and fix or remove
2. **Hidden missing value** — a sentinel code (9999, -99) standing in for "no data" → replace with `NA` using `na_if()` (see Step 3)
3. **Real extreme observation** — a genuinely large county, an unusually hot day → keep it, but run your analysis with and without it to check whether it drives your results

Step 5 — Distribution Shape

Histograms: The Workhorse

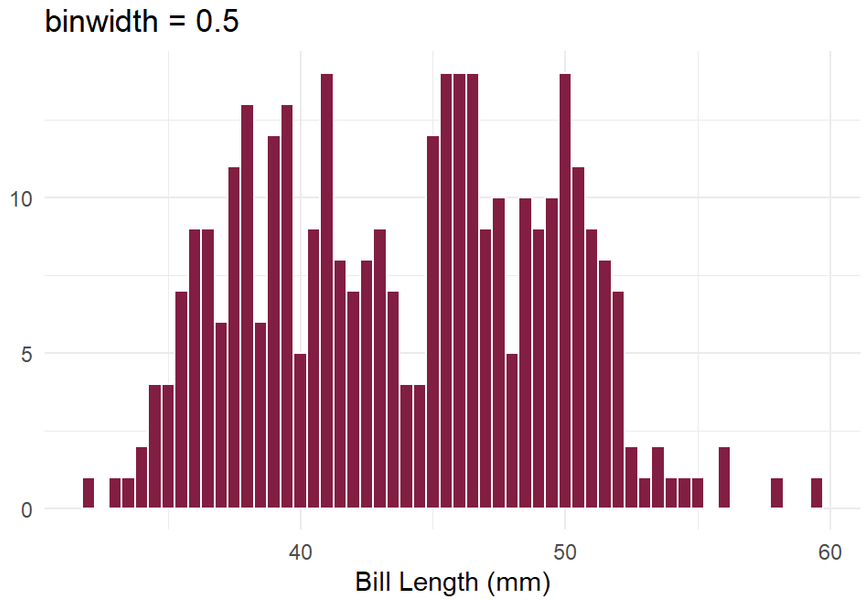
```
penguins |>
  ggplot(aes(x = bill_length_mm)) +
  geom_histogram(binwidth = 2, fill = "#861F41", color = "white") +
  labs(
    title = "Distribution of Penguin Bill Length",
    x     = "Bill Length (mm)",
    y     = "Count"
  ) +
  theme_minimal(base_size = 13)
```

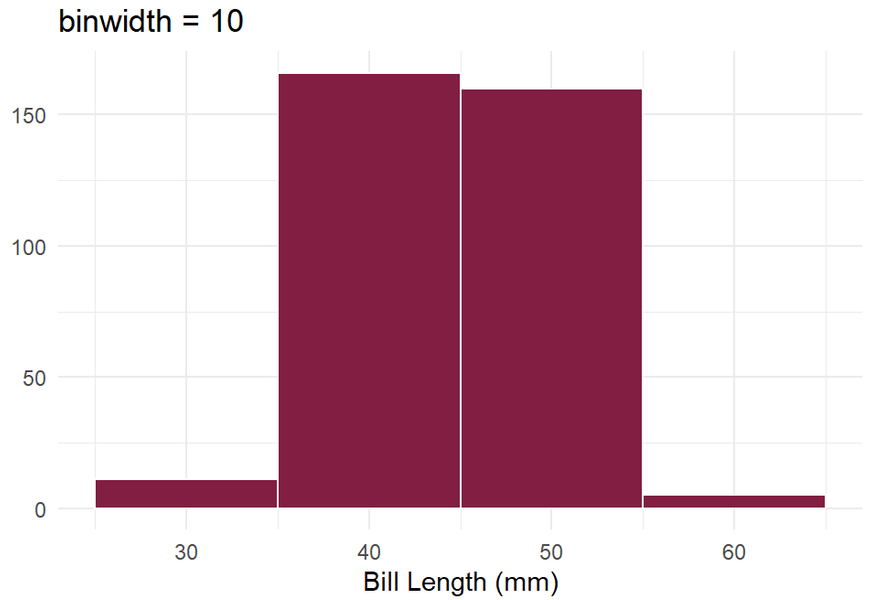


Choosing `binwidth` Matters

```
penguins |>
  ggplot(aes(x = bill_length_mm)) +
  geom_histogram(binwidth = 0.5,
    fill = "#861F41", color = "white") +
  labs(title = "binwidth = 0.5",
       x = "Bill Length (mm)", y = NULL) +
  theme_minimal()
```

```
penguins |>
  ggplot(aes(x = bill_length_mm)) +
  geom_histogram(binwidth = 10,
    fill = "#861F41", color = "white") +
  labs(title = "binwidth = 10",
       x = "Bill Length (mm)", y = NULL) +
  theme_minimal()
```





Too narrow = noise. Too wide = hides structure. Try several values before settling.

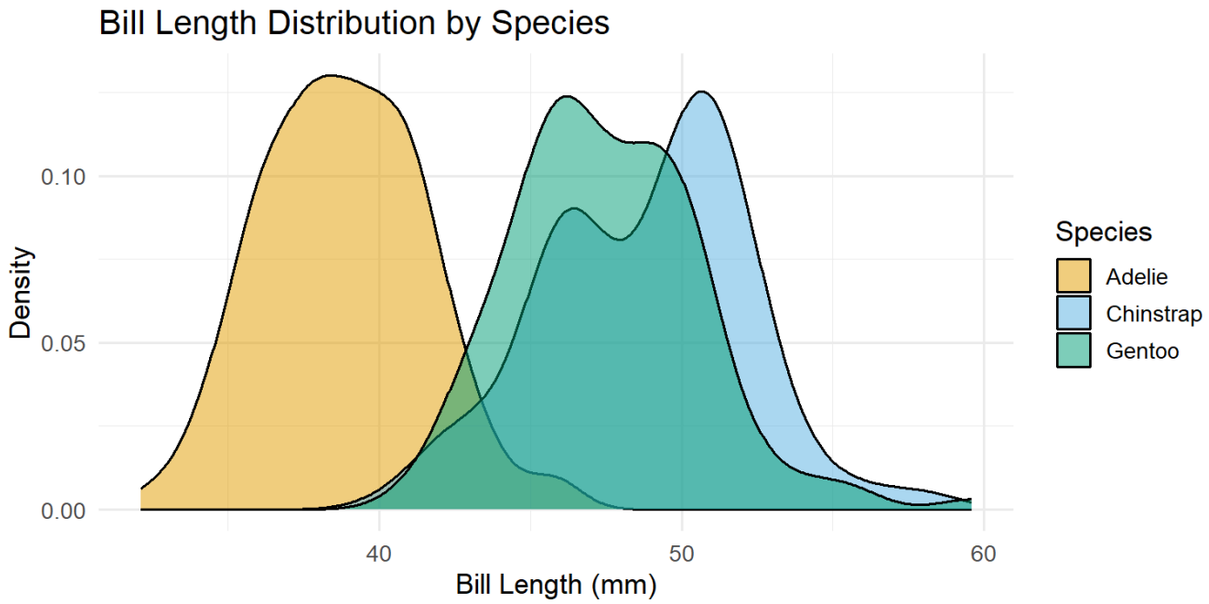
Skewness and the Log Transformation

```
p1 <- ggplot(state_data, aes(x = median_income)) +
  geom_histogram(bins = 20, fill = "#861F41", color = "white") +
  scale_x_continuous(labels = label_dollar(scale = 1e-3, suffix = "k")) +
  labs(title = "Raw: right-skewed",
       x = "Median Income", y = "Count") +
  theme_minimal(base_size = 13)

p2 <- ggplot(state_data, aes(x = log(median_income))) +
  geom_histogram(bins = 20, fill = "#E5751F", color = "white") +
  labs(title = "Log: more symmetric",
       x = "log(Median Income)", y = "Count") +
  theme_minimal(base_size = 13)

p1 + p2
```

When to use log transformation:

Density Plots: Smooth the Story

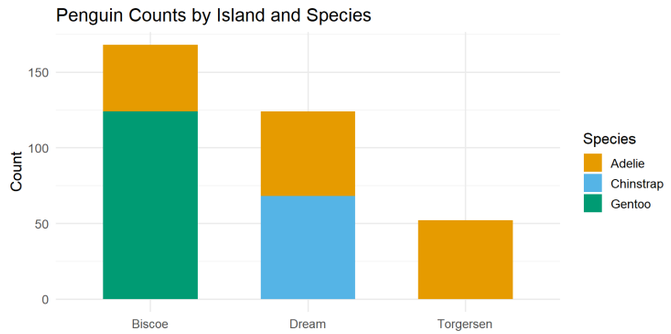
```r
penguins |>
  ggplot(aes(x = bill_length_mm, fill = species)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("#E69F00", "#56B4E9", "#009E73")) +
  labs(
    title = "Bill Length Distribution by Species",
    x     = "Bill Length (mm)",
    y     = "Density",
    fill  = "Species"
  ) +
  theme_minimal(base_size = 13)
```



Bar Charts: Discrete Variable Distribution

For **categorical or discrete** variables, use a bar chart instead:

```
penguins |>
  ggplot(aes(x = island, fill = species)) +
  geom_bar(width = 0.6) +
  scale_fill_manual(values = c("#E69F00", "#56B4E9", "#009E73")) +
  labs(
    title = "Penguin Counts by Island and Species",
    x     = NULL,
    y     = "Count",
    fill  = "Species"
  ) +
  theme_minimal(base_size = 13)
```



Histogram → continuous variable (bill length, income, temperature)

Bar chart → categorical or discrete variable (species, island, county, year)

Step 6 — Correlations

Why Look at Correlations?

After understanding each variable individually, look at how they relate. Four reasons:

1. Find your key relationships

Which variables move with your outcome? A correlation matrix gives a quick screen before formal modeling.

2. Check for multicollinearity

If two *predictors* are highly correlated (|r| > 0.7–0.8), including both in OLS inflates standard errors and makes coefficients unstable.

```
# Rule of thumb:
# |r| > 0.7 between two predictors → investigate
# Keep one, or use a combined index
```

3. Sanity check

Known relationships should appear: income and education positively correlated, poverty and income negatively correlated. If they don't, something is wrong with your data.

4. Spot unexpected patterns

An unexpected strong correlation may indicate a confounding variable you haven't controlled for — or a data problem (merge error creating spurious correlation).

> Correlation ≠ causation — but a missing expected correlation is a red flag just as much as a surprising one.

Are Variables Related?

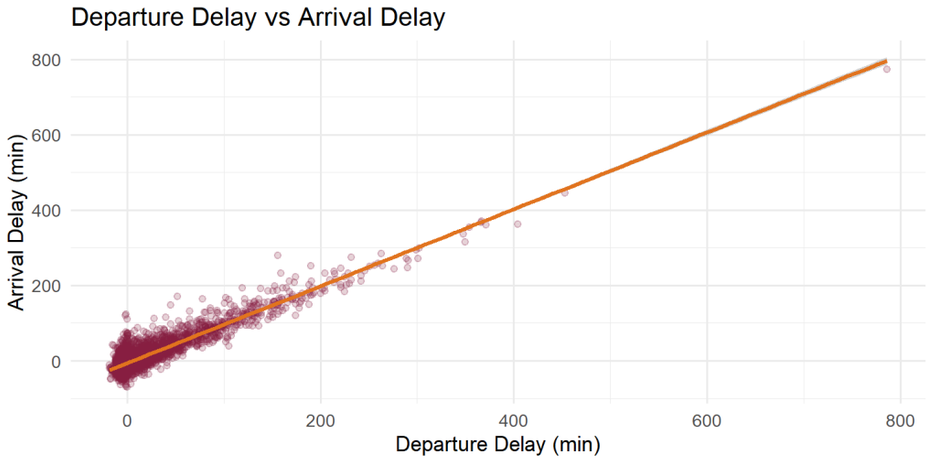
We will use **`nycflights13::flights`** — 336,776 flights from NYC airports in 2013.

```
flights |>
  select(dep_delay, arr_delay, air_time, distance) |>
  glimpse()
```

```
Rows: 336,776
Columns: 4
$ dep_delay <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -1, 0, …
$ arr_delay <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -14, 31…
$ air_time  <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 158, 3…
$ distance  <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, 1028,…
```

Scatterplot with a Trend Line

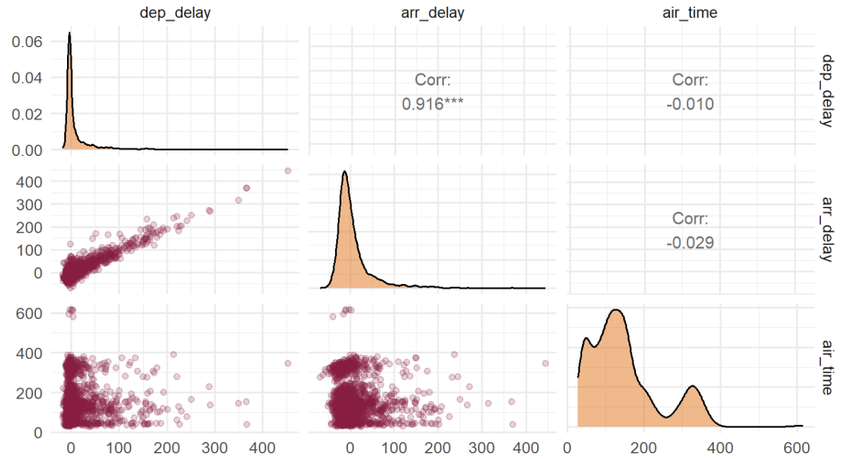
```r
set.seed(42)
flights |>
  slice_sample(n = 5000) |>
  ggplot(aes(x = dep_delay, y = arr_delay)) +
  geom_point(alpha = 0.2, color = "#861F41") +
  geom_smooth(method = "lm", color = "#E5751F", se = TRUE) +
  labs(
    title = "Departure Delay vs Arrival Delay",
    x     = "Departure Delay (min)",
    y     = "Arrival Delay (min)"
  ) +
  theme_minimal(base_size = 13)
```



`geom_smooth(method = "lm")` adds a linear regression line with confidence band.

Scatterplot Matrix: `GGally::ggpairs()`

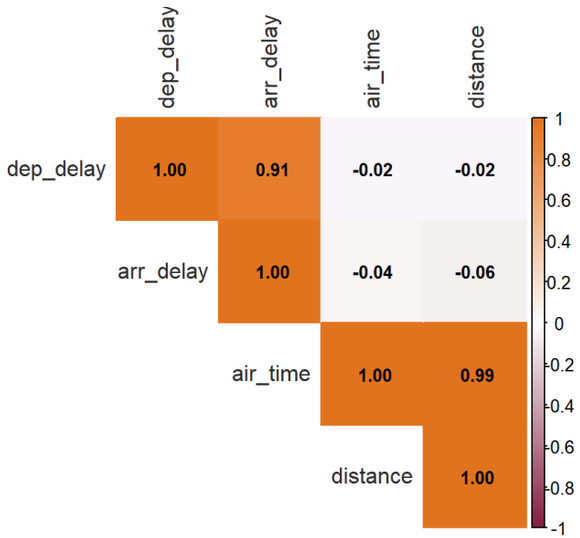
```
set.seed(42)
flights |>
  slice_sample(n = 2000) |>
  select(dep_delay, arr_delay, air_time) |>
  drop_na() |>
  ggpairs(
    lower = list(continuous = wrap("points", alpha = 0.2, color = "#861F41")),
    upper = list(continuous = wrap("cor", size = 4)),
    diag  = list(continuous = wrap("densityDiag", fill = "#E5751F", alpha = 0.5))
  ) +
  theme_minimal(base_size = 13)
```



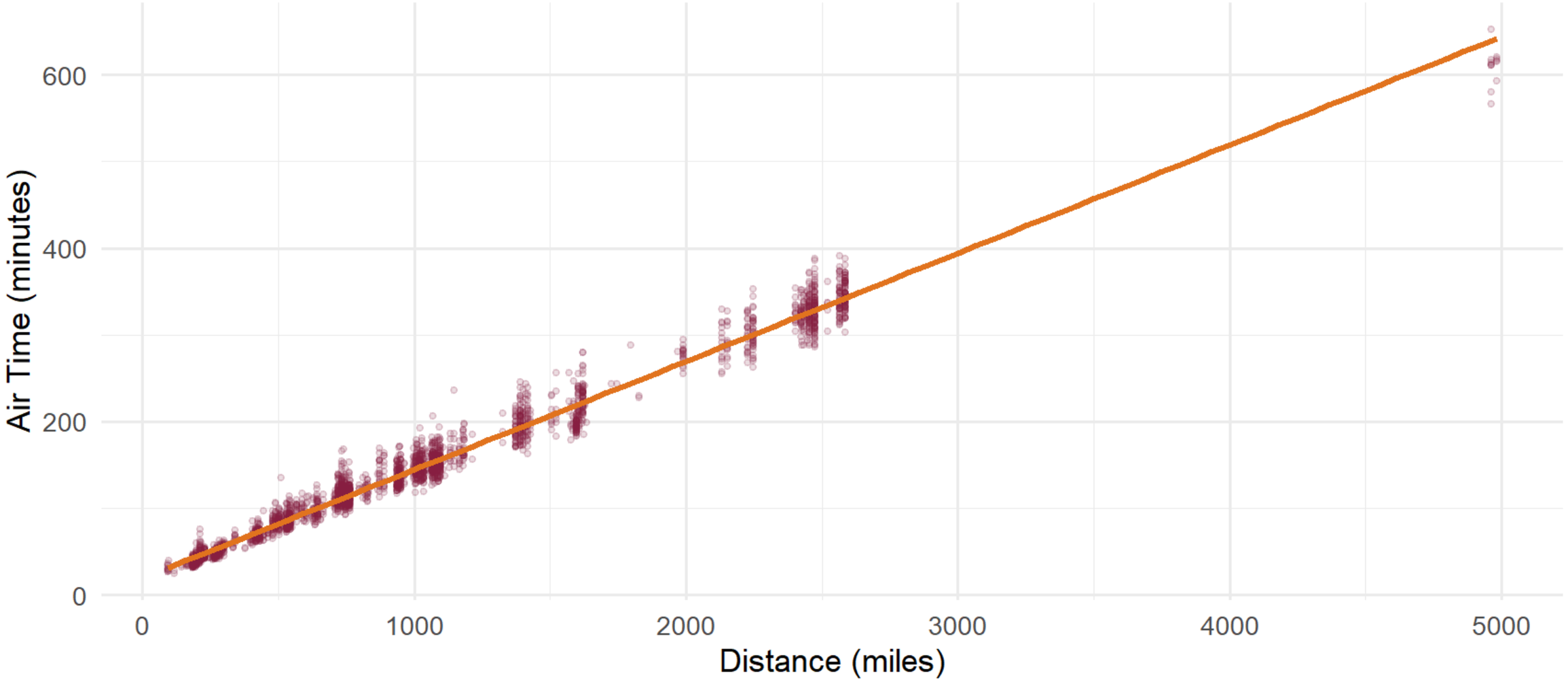
→ Open `R/demos/dataviz02/01-eda-workflow.R` (Step 6) to run `ggpairs()` and `corrplot()` on the flights data.

Correlation Heatmap: `corrplot`

```r
flights_num <- flights |>
  select(dep_delay, arr_delay, air_time, distance) |>
  drop_na()

corrplot(
  cor(flights_num),
  method     = "color",
  type       = "upper",
  col        = colorRampPalette(c("#861F41", "white", "#E5751F"))(200),
  tl.col     = "#2D2D2D",
  addCoef.col = "black",
  number.cex = 0.8
)
```



🔍 Describe This Graph in One Sentence



Answer

> **“Flight distance and air time are almost perfectly linearly related — longer routes take proportionally longer to fly.”**

Research implication: If your model includes both `distance` and `air_time` as predictors, this near-perfect correlation (r ≈ 0.99) means severe multicollinearity — you should include only one.

Which Plot?

Same Data, Four Charts — Comparing Groups

Scenario: You want to compare the distribution of flipper length across 3 penguin species.

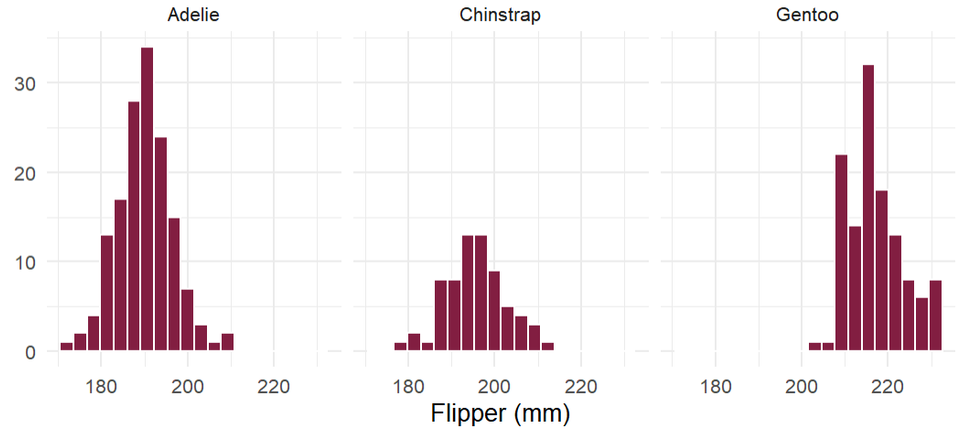
Which chart type best shows how the three species differ?

Option	Chart type
A	Faceted histogram
B	Overlapping density
C	Violin + jitter
D	Line chart

Same Data, Four Charts — Comparing Groups (Answer)

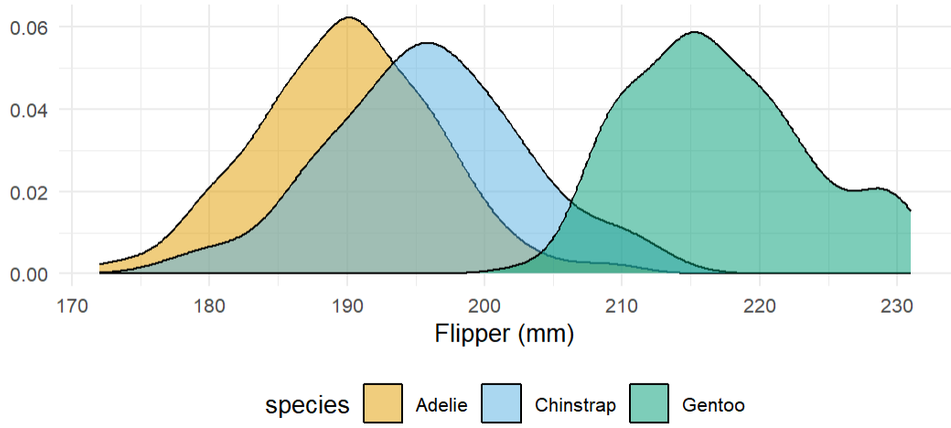
Scenario: You want to compare the distribution of flipper length across 3 penguin species.

A: Faceted histogram



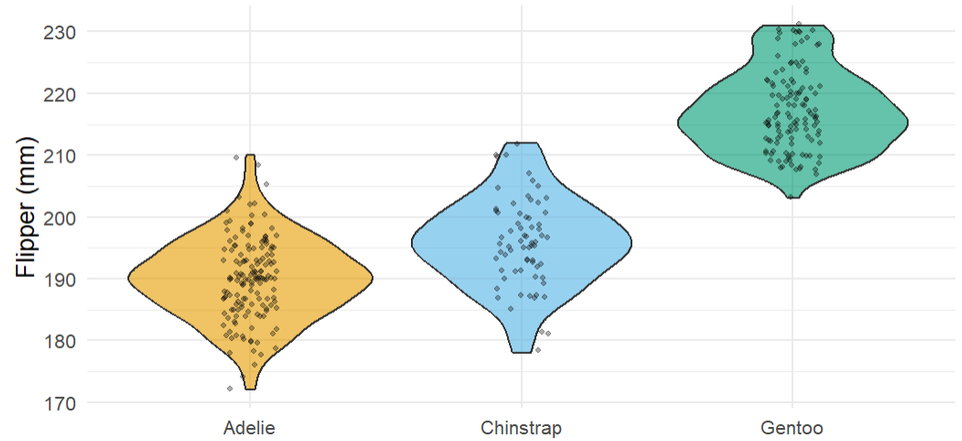
Works — but eye must travel across panels to compare.

B: Overlapping density ✓



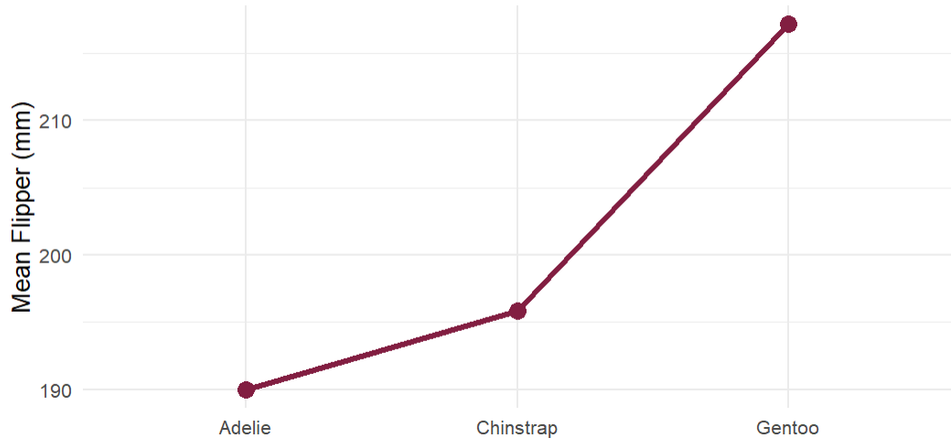
Good — direct comparison, shape visible.

C: Violin + jitter ✓✓



Best. Shape + individual points + easy comparison.

D: Line chart ✗



Wrong tool. Lines imply sequence/order — species are unordered categories.

Same Data, Four Charts — Two Continuous Variables

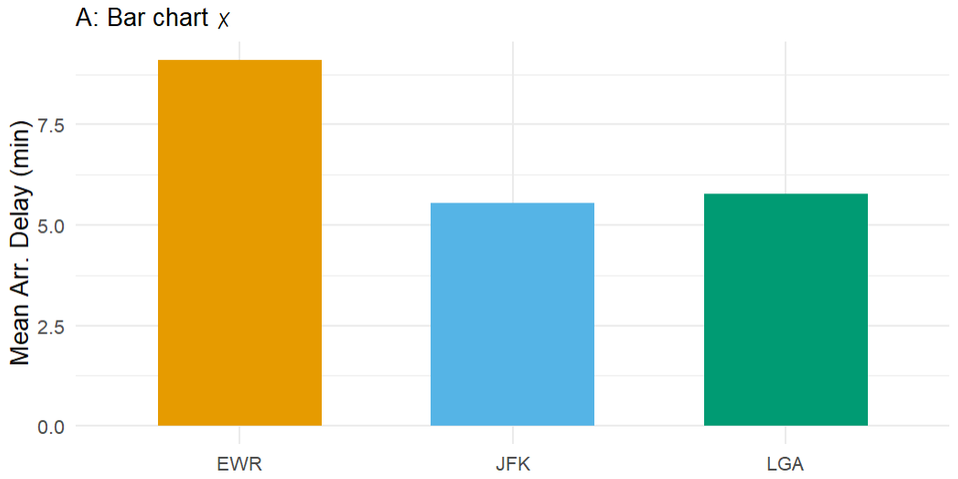
Scenario: You want to understand the relationship between departure delay and arrival delay.

Which chart type best reveals this relationship?

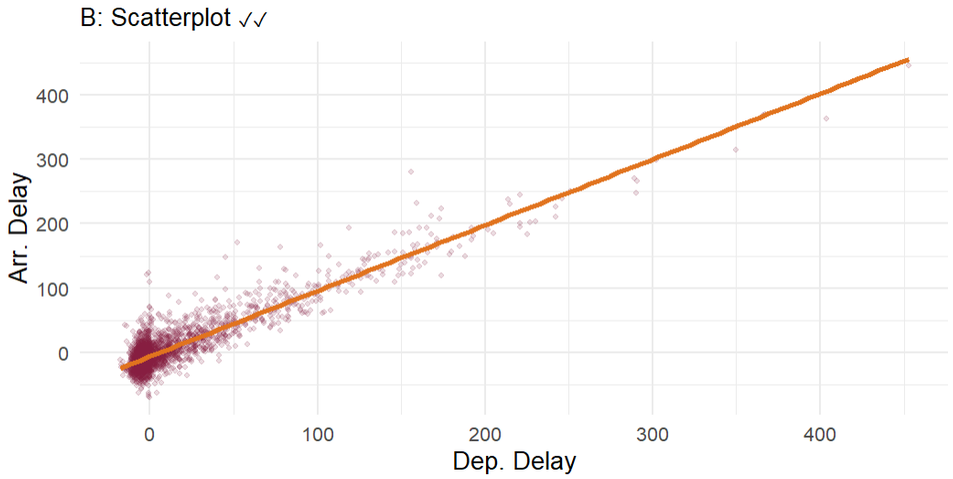
Option	Chart type
A	Bar chart (mean by airport)
B	Scatterplot with trend line
C	Box plot by airport
D	Histogram of arrival delay

Same Data, Four Charts — Two Continuous Variables (Answer)

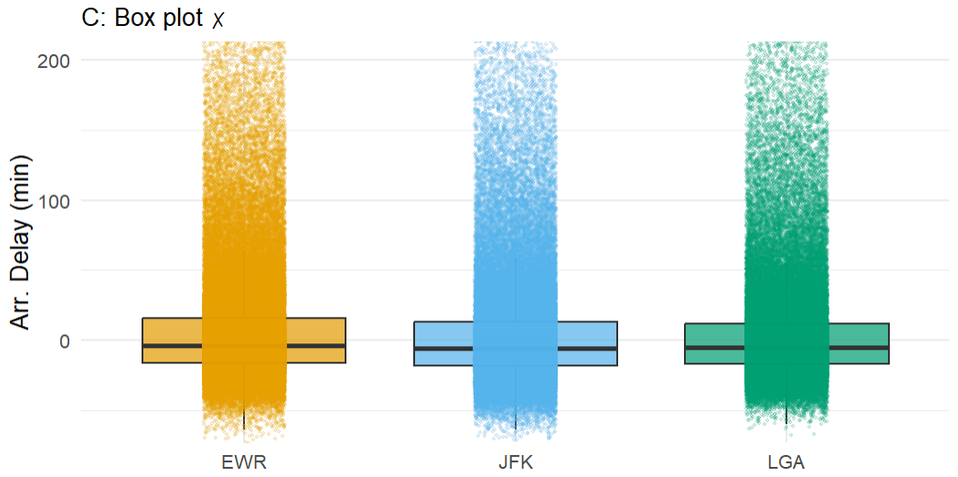
Scenario: You want to understand the relationship between departure delay and arrival delay.



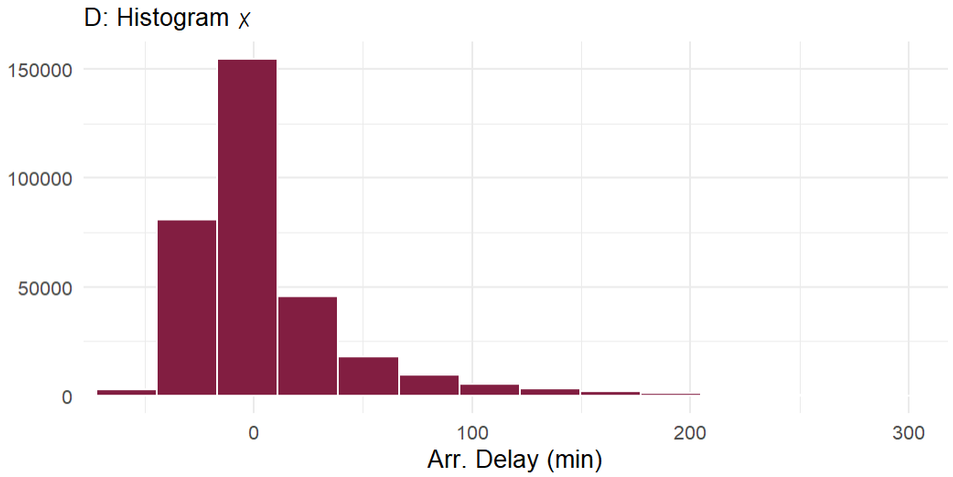
Aggregates to means — hides the variation and individual relationship.



Best. Shows relationship direction, strength, and individual variation.



Shows distribution per airport but ignores the departure delay variable entirely.



Single variable only — does not address the question about the *relationship*.

Same Data, Four Charts — Composition

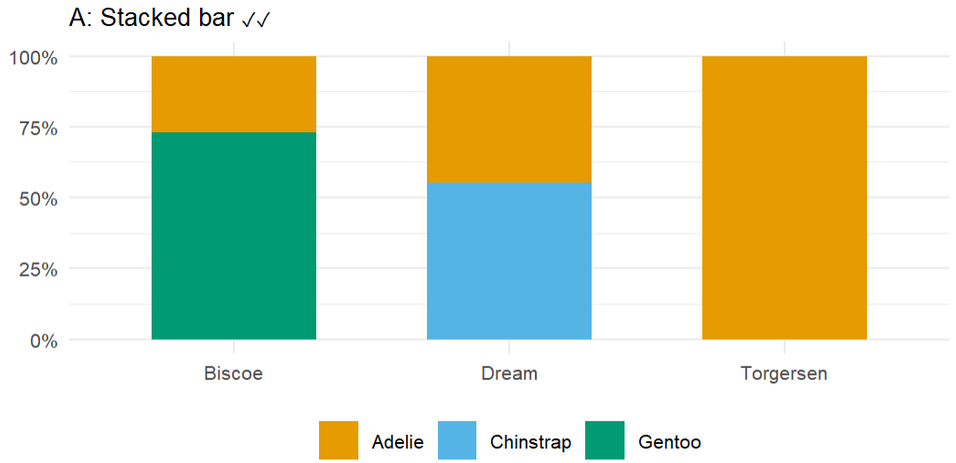
Scenario: You want to show what share of penguins from each island belong to each species.

Which chart type makes the proportions easiest to read?

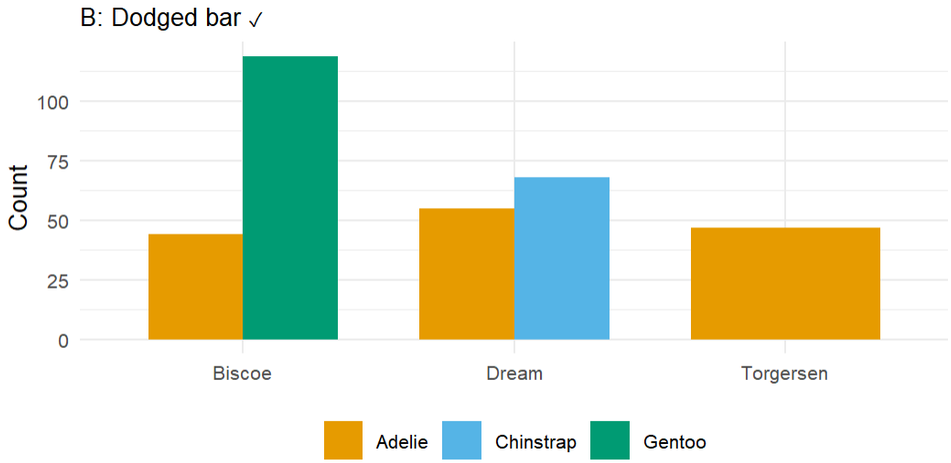
Option	Chart type
A	Stacked proportional bar
B	Dodged bar (counts)
C	Jitter plot
D	Line chart

Same Data, Four Charts — Composition (Answer)

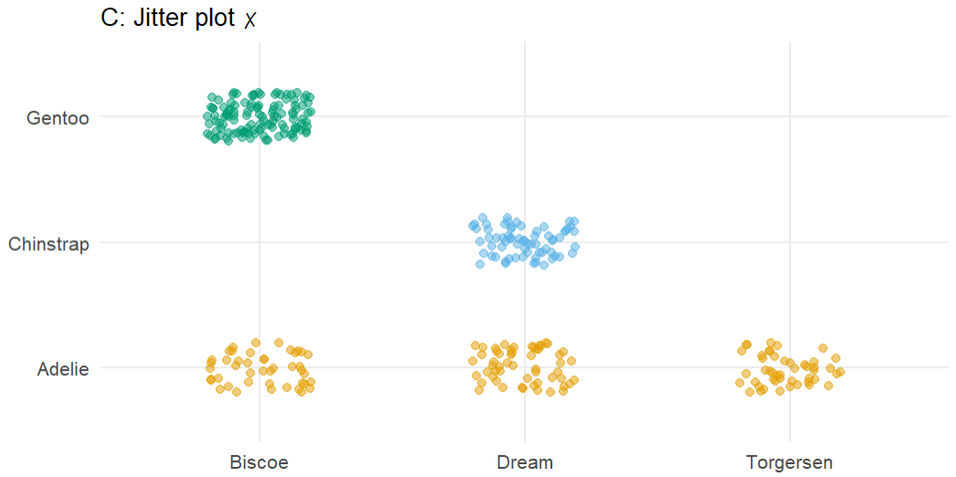
Scenario: You want to show what share of penguins from each island belong to each species.



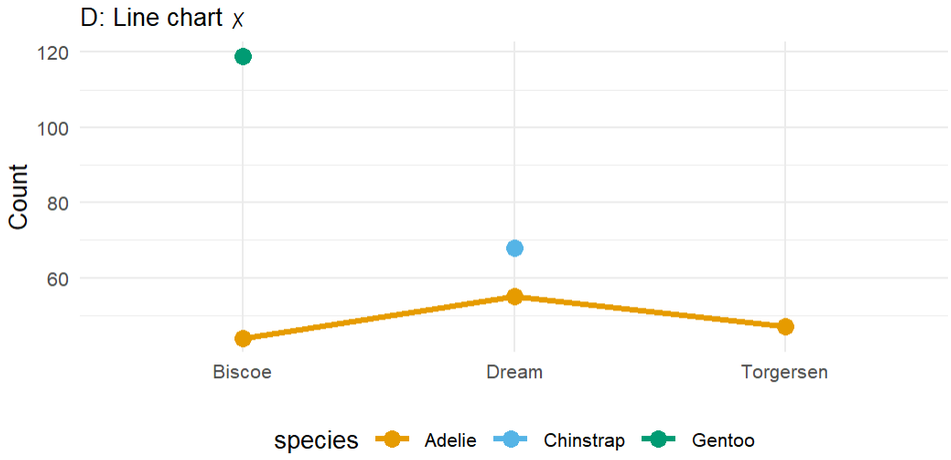
Best. Proportions are directly readable for each island.



Good for counts — harder to read proportions when groups differ in total size.



Shows individual points but obscures proportions — counting dots is hard.



Lines imply a continuous sequence — islands are not ordered on a scale.

Run the 6-step EDA workflow on your project data

```r
library(tidycensus)
library(skimr)

va_counties <- get_acs(
  geography = "county", state = "VA",
  variables = c(median_income = "B19013_001",
                pov_count     = "B17001_002",
                pov_total     = "B17001_001",
                pct_college   = "B15003_022"),
  year = 2022, output = "wide"
) |>
  mutate(pov_rate = pov_countE / pov_totalE)

# Step 1: skim(va_counties) — what does the output tell you?
# Step 2: Any n_missing > 0? Any max values that look impossible?
# Step 3: geom_boxplot of median_incomeE — any outliers?
# Step 4: geom_histogram of pov_rate — shape? skewed?
#         Try log(pov_rate) — does it look more symmetric?
# Step 5: Scatterplot pov_rate (x) vs median_incomeE (y)
#         Add geom_smooth(). In one sentence: what does this show?
```

Key Takeaways

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- `skim()` **first, always** — one function call reveals missing counts, ranges, and shape before you look at anything else
- **Check for problems before visualizing** — impossible values, sentinel codes, and logical inconsistencies corrupt downstream analysis silently
- **Not all missing is** `NA` — sentinel codes (9, 99, 9999, -99) need `na_if()`; logically missing values need domain judgment
- **Boxplot for anomalies** — investigate outliers before dropping them; they may be data errors or real extreme cases
- **Histogram for shape** — right-skewed distributions often benefit from a log transformation; check your outcome and key predictors
- **Correlations for relationships** — screen predictors against your outcome; check for multicollinearity among predictors ($|r| > 0.7 \rightarrow$ investigate)